

Name: _____
 Class: _____
 Date: _____

Mr. Rawson • GCSE Physics

REVIEW — Series Resistors

Keep this sheet. Everything you need for series circuits is on this side.

Key concept

THE THREE RULES — series (one loop, no choice of route)

1. **Current is the same everywhere.** $I_1 = I_2 = I_{\text{supply}}$

Charge is not used up. An ammeter reads the same wherever you put it in the loop.

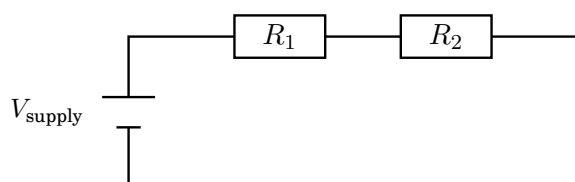
2. **Potential difference is shared out.** $V_{\text{supply}} = V_1 + V_2$

The bigger resistor takes the bigger share.

3. **Resistances add.** $R_{\text{total}} = R_1 + R_2$

Adding a resistor in series always *increases* total resistance — the charge must pass through every one of them, so there is more opposition on the single route available.

And $V = IR$ still applies to *every single component*, as well as to the whole circuit.



same current all the way round

Worked example

WORKED EXAMPLE 1 — the standard one

A $4\ \Omega$ and a $6\ \Omega$ resistor are connected in series to a $12\ \text{V}$ supply. Find the current, and the potential difference across each resistor.

1. **Total resistance** $R_{\text{total}} = R_1 + R_2 = 4 + 6 = 10\ \Omega$
2. **Current** $I = \frac{V}{R} = \frac{12}{10} = 1.2\ \text{A}$ (this is the current *everywhere*)
3. **Each p.d.** $V_1 = IR_1 = 1.2 \times 4 = 4.8\ \text{V}$
 $V_2 = IR_2 = 1.2 \times 6 = 7.2\ \text{V}$
4. **Check** $4.8 + 7.2 = 12\ \text{V}$ ✓ matches the supply — rule 2

Worked example

WORKED EXAMPLE 2 — working backwards

A $9\ \text{V}$ supply drives a current of $0.30\ \text{A}$ through two resistors in series. One of them is $12\ \Omega$. Find the other.

1. **Total resistance** $R_{\text{total}} = \frac{V}{I} = \frac{9}{0.30} = 30\ \Omega$
2. **Subtract** $R_2 = R_{\text{total}} - R_1 = 30 - 12 = 18\ \Omega$

REVIEW — Series Resistors: practice

- 1) **State** what happens to the size of the current as you move an ammeter to different points around a series circuit. [1]

It stays the same — the current is identical at every point in a series loop.

- 2) **Calculate** the total resistance of a $3\ \Omega$ and a $7\ \Omega$ resistor in series. [1]

$$R_{total} = 3 + 7 = 10\ \Omega$$

- 3) **Calculate** the current when those two resistors are connected to a 20 V supply. [2]

$$I = V/R = 20/10 = 2.0\text{ A}$$

- 4) **Calculate** the potential difference across each of the two resistors. [2]

$$V_1 = IR_1 = 2.0 \times 3 = 6.0\text{ V} \quad \text{and} \quad V_2 = IR_2 = 2.0 \times 7 = 14\text{ V}$$

- 5) **Show** that your two answers above are consistent with the supply potential difference. [1]

$6.0 + 14 = 20\text{ V}$, which equals the supply p.d. — the p.d. is shared between components.

- 6) **Calculate** the unknown resistance: a 12 V supply drives 0.40 A through two resistors in series, one of which is $10\ \Omega$. [3]

$$R_{total} = V/I = 12/0.40 = 30\ \Omega; \quad R_2 = 30 - 10 = 20\ \Omega$$

- 7) **Explain** qualitatively why adding another resistor in series *increases* the total resistance of a circuit. [2]

There is only one route, so the charge must pass through every resistor in turn. Each one opposes the flow, so the total opposition rises. The current therefore falls for the same supply p.d.

- 8) **Explain** what happens to the other lamp when one lamp in a series circuit is removed, and why. [2]

The other lamp goes out. There is a single loop, so removing one component breaks the only route and the current stops everywhere.